Paris Lodron University Salzburg

Research-Based Work Placement Report

INT_OPT-Collaborative research/work placements

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1. Introduction

The rapid grown of geospatial artificial intelligence (GeoAI) is unfolding new capabilities in the geospatial domain as never before. It's not only transforming already existing methodologies and making space for new outcomes to come in geospatial information systems (GIS) field, but it's also revolutionizing the way decision-making is performed in multiple industries (Li, 2025; Li & Hsu, 2022; Döllner, 2020). At the same time, regulations like the EU Deforestation Regulation (EUDR) are reshaping global supply chains. While essential for sustainability, these regulations cause substantial challenges for smallholder farmers, who often lack the ability and technical capacity to verify geolocation, assess risk, and demonstrate compliance (Judijanto, 2025; Melati & Jintarith, 2024).

The aim of the thesis is to develop a comprehensive GeoAI framework that bridges the gap between EUDR requirements and the reality of smallholder farmers, proving its applicability and providing value beyond the regulation. In order to achieve this, the project will be divided into two parts. The first part introduces a theoretical GeoAI framework driven by the exponential expansion and its increasing relevance across industries. This framework explains how GeoAI improves accessibility, scalability, automation, decision-making, and transparency in the geospatial domain, establishing the conceptual background for the project.

The second part applies this framework to a real-world case study, which will focus on enabling Colombian small coffee farmers, in the Quindío region, in complying with EUDR requirements. The project relies on plot-level geolocated farm data, consisting of spatial polygons representing individual coffee farm plots. These data enable the validation of farm geolocation and support the generation of Earth Observation indicators required for EUDR compliance.

To address some of the main challenges small farmers face, this thesis will develop a replicable pipeline using GeoAI and implementing primarily in Python that automates geolocation validation of their plots, integrates models such as Sentinel-2 Deep Resolution 3.0 (S2DR3) and Segmented Anything for Earth Observation (SAM4EO), and enriches assessments with meaningful Earth Observation indicators.

As a result, an interactive dashboard developed in JavaScript that presents farmers with clear EUDR compliance results and added-value insights about their plots will be delivered. This prototype demonstrates the practical use of the GeoAI framework and serves as a scalable blueprint for future applications beyond the case study.

2. Motivation

This collaborative project initiative is the continuation of the research conducted during the internship at GeoCitizens, where cutting-edge technologies were explored as benchmarking for the small farmers EUDR compliance. Despite achieving the intended objectives during the exploration of these tools, time constraints represented a challenge to translating these tools into practice. Leading to the opportunity to expand the project to a long-lasting higher impact research.

3. Objectives

1. Develop a theoretical GeoAI framework that explains the role of GeoAI in regulatory and sustainability-focused environments, highlighting its contribution to accessibility, scalability, automation, decision support, and transparency in geospatial applications.
2. Design and implement a scalable GeoAI-based compliance pipeline that applies the proposed framework to a real-world case study, supporting Colombian smallholder coffee and cacao farmers in meeting EUDR requirements.
3. Translate the GeoAI pipeline into an interactive, user-centered dashboard that presents EUDR compliance outcomes together with value-added, plot-level geospatial insights in an accessible and interpretable manner.
4. Ensure methodological transferability and scientific rigor by documenting the pipeline architecture, methodology, and assessment criteria at an abstract level, enabling the case study to serve as a blueprint for future GeoAI-based compliance solutions while protecting proprietary interests.

4. Methodology

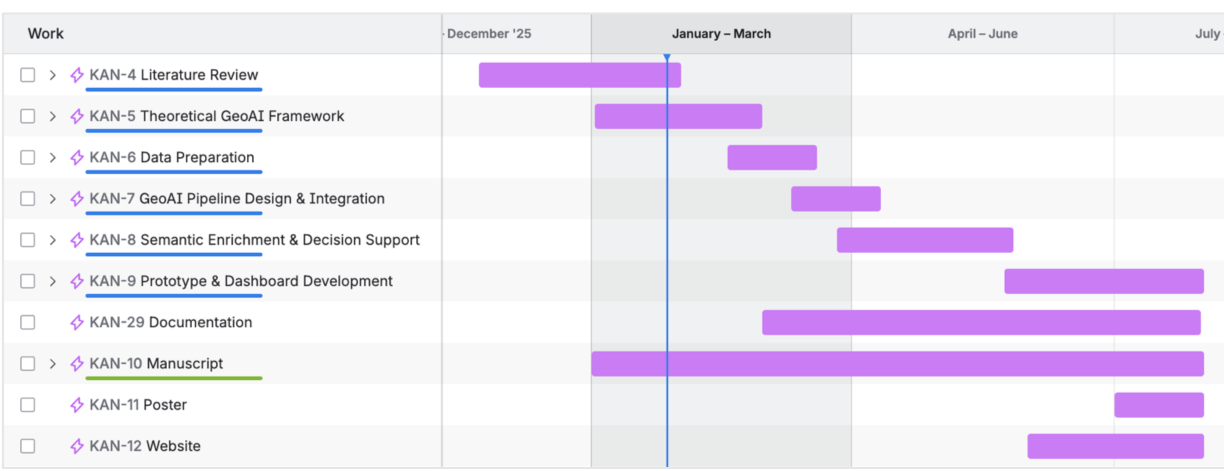

Figure 1. Gannt Chart

As illustrates in Figure 1, the project is divided into 10 tasks, which at the same time can be categorized across three big categories. The first two phases cover the project's initial phases: establishing the theoretical framework and applying it to a real-world case study. This is followed by a final phase dedicated to the production of supplementary materials for the final submission.

Specifically, tasks KAN-4 and KAN-5 represent the development of the theoretical foundation. These involve a comprehensive exploration of the thesis's core topics— GeoAI and EUDR —as well as a focused framing of the case study. Tasks KAN-7, 8, and 9 constitute the application and implementation stages, including data preparation, GeoAI model design, semantic acquisition, and dashboard implementation. Finally, tasks KAN-10, 11, and 12 encompass the complementary work required for the dissemination and final submission of the master's thesis.

5. Expected Outcomes

Regarding the final outputs, this thesis aims to produce two complementary results that directly relate to the two major phases of the project. The first consists in a theoretical GeoAI framework that exposes the role of GeoAI regulatory and sustainability-focused environments. This framework will outline how GeoAI contributes to improved accessibility, scalability, automation, decision support, and transparency in geospatial applications.

Parallelly, the GeoAI foundation will be applied and validated through a real-world case study in collaboration with GeoCitizens. The second final output will be a scalable GeoAI-based compliance pipeline focused on supporting Colombian small coffee and cacao farmers in meeting EUDR compliance. The solution will be presented as an interactive dashboard that will show compliance outcomes as well as value-added information at the plot level.

Although the case study is project-specific to the collaborative environment, the methodology, architecture, and assessment criteria will be presented at an abstract level. This will ensure scientific rigor while at the same time protecting intellectual property rights, thereby allowing the case study to act as a blueprint for GeoAI-based compliance solutions beyond the scope of the case study.

6. References

Andrei, N., & Scarlat, C. (2023, December). *An overview of AI and geospatial data towards improved strategic decisions and automated business decision processes*. In International Conference of Management and Industrial Engineering (Vol. 11, pp. 161–168). https://doi.org/10.56177/11icmie2023.30

Cao, K., Zhou, C., Church, R., Li, X., & Li, W. (2024). Revisiting spatial optimization in the era of geospatial big data and GeoAI. *International Journal of Applied Earth Observation and Geoinformation, 129*, 103832. https://doi.org/10.1016/j.jag.2024.103832

Corona, P., Di Stefano, V., & Mariano, A. (2023). Knowledge gaps and research opportunities in the light of the European Union regulation on deforestation-free products. *Annals of Silvicultural Research, 48*, 87–89. https://doi.org/10.12899/asr-2445

Dold, J., & Groopman, J. (2017). The future of geospatial intelligence. *Geospatial Information Science, 20*(2), 151–162. https://doi.org/10.1080/10095020.2017.1337318

Döllner, J. (2020). Geospatial artificial intelligence: Potentials of machine learning for 3D point clouds and geospatial digital twins. *PFG – Journal of Photogrammetry, Remote Sensing and Geoinformation Science, 88*(1), 15–24. https://doi.org/10.1007/s41064-020-00102-3

Fisher, M. R., Obidzinski, K., Mota Alves, A., & Ekaputri, A. D. (2024). *Commodities and global climate governance: Early evidence from the EU deforestation-free regulation (EUDR).* https://hdl.handle.net/10125/110279

Freitas Beyer, J., Köthke, M., & Lippe, M. (2025). Assessing the suitability of available global forest maps as reference tools for EUDR-compliant deforestation monitoring. *Remote Sensing, 17*(17), 3012. https://doi.org/10.3390/rs17173012

Judijanto, L. (2026). Who pays for climate action? The disproportionate impact of deforestation-free regulations on smallholder farmers. *European Journal of Management, Economics and Business, 3*(1), 3–13. https://doi.org/10.59324/ejmeb.2026.3(1).01

Li, B. (2025). From maps to decisions: A GeoAI framework for multi-hazard infrastructure resilience and equitable emergency management. *American Journal of Big Data, 6*(3), 139–153. https://doi.org/10.71465/ajbd3432

Li, W., & Hsu, C. Y. (2022). GeoAI for large-scale image analysis and machine vision: Recent progress of artificial intelligence in geography. *ISPRS International Journal of Geo-Information, 11*(7), 385. https://doi.org/10.3390/ijgi11070385

Li, W., Arundel, S., Gao, S., Goodchild, M., Hu, Y., Wang, S., & Zipf, A. (2024). GeoAI for science and the science of GeoAI. *Journal of Spatial Information Science*, (29), 1–17. https://doi.org/10.5311/JOSIS.2024.29.349

Liu, P., & Biljecki, F. (2022). A review of spatially explicit GeoAI applications in urban geography. *International Journal of Applied Earth Observation and Geoinformation, 112*, 102936.

Melati, K., & Jintarith, P. (2024). *Finding a place for smallholder farmers in EU deforestation regulation.* https://doi.org/10.51414/sei2024.035

Rothe, P., Celik Schoreels, I., Phuntsho, P., & Henke, J. (2024, July). *EUDR coffee check: A tool to support the coffee sector to become compliant with the European Union regulation on deforestation-free products (EUDR).* In Proceedings (Vol. 109, No. 1, p. 21). MDPI. https://doi.org/10.3390/ICC2024-18160

UNESCO. (2022). *Recommendation on the ethics of artificial intelligence.* United Nations Educational, Scientific and Cultural Organization.

van Noordwijk, M., Dewi, S., Minang, P. A., Harrison, R. D., Leimona, B., Ekadinata, A., … Sayer, J. (2025). Beyond imperfect maps: Evidence for EUDR-compliant agroforestry. *People and Nature.* https://doi.org/10.1002/pan3.70088

Wang, S., Huang, X., Liu, P., Zhang, M., Biljecki, F., Hu, T., … Bao, S. (2024). Mapping the landscape and roadmap of geospatial artificial intelligence (GeoAI) in quantitative human geography: An extensive systematic review. *International Journal of Applied Earth Observation and Geoinformation, 128*, 103734. https://doi.org/10.1016/j.jag.2024.103734